

creditlib

A Single-Name CDS Pricing Engine

Derivations, Implementation, Validation, and Application

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Prepared as a portfolio project for credit / fixed-income analyst roles

Scope. A single-name credit default swap pricing engine built from first principles in Python. Every formula is derived before it is coded. Every component is validated against no-arbitrage conditions and, where a comparable object exists, against QuantLib. The library bootstraps a hazard-rate term structure from quoted CDS spreads, prices both legs of a CDS by exact closed-form integration, and produces the standard credit risk measures.

Status. Steps 1–5 complete. 86 tests passing. Applied to a five-name U.S. bank holding company cross-section.

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1 Overview and Design Philosophy

1.1 What the library does

`creditlib` prices single-name credit default swaps. Given a discount curve and a survival (credit) curve it computes the present value of both legs, the par spread, the mark-to-market, and the standard risk measures. Run in reverse, it bootstraps the survival curve from quoted market spreads.

The stack is Python with NumPy, SciPy, pandas and matplotlib. QuantLib is used exclusively as an independent check — never as a dependency of the pricing path.

1.2 Two rules that shaped every decision

1. **Derive before coding.** No formula is asserted. Every expression in this document is obtained from a stated definition or a replication argument, so that each line of the implementation can be defended.
2. **Validate against no-arbitrage and against QuantLib.** Structural conditions (positivity, monotonicity, $Q(0) = 1$) are enforced by construction where possible and asserted at runtime where not. Numerical agreement with QuantLib is asserted at the tightest tolerance the conventions permit, and where conventions differ the discrepancy is explained and quantified rather than absorbed into a loose tolerance.

1.3 The design decision that made everything else work

Both objects the pricer needs — a discount curve and a survival curve — have the same mathematical form:

$$\text{value}(t) = \exp\left(-\int_0^t \text{rate}(s) ds\right)$$

with a *piecewise-constant* instantaneous rate. For the discount curve that rate is the instantaneous forward $f(s)$; for the credit curve it is the hazard rate $h(s)$.

Choosing piecewise-constant rates was not a convenience. It has three consequences that each turn out to be load-bearing:

1. **No-arbitrage becomes a sign constraint.** A survival curve is admissible if and only if $h \geq 0$. This is a box constraint on a scalar rather than a shape constraint policed after the fact.
2. **The pricing integrals close in elementary form.** Because f and h are both constant on the merged grid of curve knots, the protection leg and accrual integrals have exact closed forms. No quadrature error is layered on top of the interpolation error already accepted.
3. **The inverse problem becomes square and triangular.** With knots placed at the quote maturities, n market quotes determine exactly n hazard rates, and the Jacobian is lower triangular — so the bootstrap is n one-dimensional root-finds rather than an n -dimensional nonlinear solve.

Consequences (2) and (3) are why the same interpolation choice appears in both curve classes. Had the discount curve used a spline on zero rates, the protection-leg integral would have required numerical quadrature and the bootstrap would have lost its clean structure.

1.4 Repository layout

```
creditlib/  
  daycount.py      ACT/360, ACT/365F, 30/360  
  curves.py        PiecewiseFlatCurve -> CreditCurve  
  discount_curve.py DiscountCurve (log-linear on discount factors)  
  schedule.py      IMM roll dates, ACT/360 accruals, seasoned stubs  
  cds.py           legs, price_cds, par_spread, CS01 / IR DV01 / Rec01 / JTD  
  bootstrap.py     survival curve from quoted spreads  
  validation.py    runtime no-arbitrage assertions  
tests/             79 tests  
notebooks/         bank_credit_analysis.ipynb
```

2 Survival Curves and Hazard Rates

2.1 Setting up

Let $\tau \geq 0$ be the default time of the reference entity, a random variable under a risk-neutral measure. The **survival function** is

$$Q(t) = \mathbb{P}(\tau > t), \quad Q(0) = 1.$$

Assume Q is absolutely continuous with density $f(t) = -Q'(t)$.

2.2 Deriving the hazard rate

The hazard rate is the *conditional* instantaneous default intensity — the rate of default over the next instant, given survival to now:

$$h(t) = \lim_{\Delta \downarrow 0} \frac{1}{\Delta} \mathbb{P}(t < \tau \leq t + \Delta \mid \tau > t).$$

Expanding the conditional probability, and noting $\{t < \tau \leq t + \Delta\} \subset \{\tau > t\}$ so the intersection is the numerator event:

$$h(t) = \lim_{\Delta \downarrow 0} \frac{1}{\Delta} \cdot \frac{Q(t) - Q(t + \Delta)}{Q(t)} = \frac{-Q'(t)}{Q(t)} = -\frac{d}{dt} \ln Q(t).$$

The hazard rate *is* the negative log-derivative of survival. Integrating from 0 to t with $Q(0) = 1$:

$$\boxed{Q(t) = \exp\left(-\int_0^t h(s) ds\right) = e^{-H(t)}} \quad (1)$$

where $H(t) = \int_0^t h(s) ds$ is the **cumulative hazard**.

2.3 Two properties worth stating out loud

h is an intensity, not a probability. It carries units of 1/time and may exceed 1. A name with $h = 3.0$ is not mis-specified; it has an expected time to default of four months.

Survival probabilities multiply. $Q(t_2)/Q(t_1)$ is the probability of surviving from t_1 to t_2 *given* survival to t_1 . This conditional structure is what makes the bootstrap sequential: each new instrument extends the curve without disturbing what came before.

2.4 Piecewise-constant hazards

Place knots at $0 = t_0 < t_1 < \dots < t_n$ and set $h(s) = h_i$ for $s \in (t_{i-1}, t_i]$. Then

$$H(t_j) = \sum_{i=1}^j h_i (t_i - t_{i-1}), \quad H(t) = H(t_{j-1}) + h_j (t - t_{j-1}) \text{ for } t \in (t_{j-1}, t_j].$$

H is continuous and piecewise linear, so $\ln Q$ is piecewise linear. Therefore:

Piecewise-constant forward hazards and log-linear interpolation of survival probabilities are the same assumption, not two competing ones. This is the ISDA CDS Standard Model convention and matches QuantLib's `BackwardFlat` interpolation.

2.5 Why parameterise in hazard space

One could bootstrap Q directly. With $Q = e^{-H}$ the admissibility conditions become:

Condition	Enforced by
$Q(0) = 1$	$H(0) = 0$, structural
$0 < Q(t) \leq 1$	$H \geq 0$, structural
Q non-increasing	$h_i \geq 0$ — one box constraint

A rising survival curve implies $\mathbb{P}(\text{default in } (t_1, t_2]) < 0$, a genuine arbitrage. In hazard space that is a sign constraint on a scalar; in survival space it is a shape constraint requiring post-hoc policing. Accordingly `CreditCurve.__init__` rejects negative rates, and `from_survival_probabilities` on a rising curve fails at *construction*, not at pricing.

2.6 Validation

- Flat hazard reproduces e^{-ht} to 10^{-15} .
- $H(t) = -\ln Q(t)$ to machine zero.
- Segment slopes of H recover the input hazards to 10^{-12} relative.
- Survival composes multiplicatively across sub-periods.
- Negative hazard, and a *rising* survival input, both rejected at construction.
- **QuantLib:** survival probabilities match `ql.HazardRateCurve` to 10^{-12} relative at the knots *and* at interior points. Matching between knots is the meaningful part — it proves the interval convention $(t_{i-1}, t_i]$ agrees, not merely the arithmetic.

3 The Discount Curve

3.1 The discount factor is the primitive

$D(t)$ is the price today of a contract paying 1 unit with certainty at time t . Three properties, and only three:

$$D(0) = 1, \quad D(t) > 0, \quad D \text{ continuous.}$$

$D(0) = 1$ is definitional. $D(t) > 0$ is the only genuine arbitrage condition — a non-positive price for a strictly positive payoff is free money.

Nothing requires D to decrease. $D(t) > 1$ means paying 1.02 today to receive 1.00 later, which is exactly a negative rate. EUR, JPY and CHF curves traded there for years. The zero lower bound is an empirical regularity arising from cash storage and custody costs, not a no-arbitrage bound. Contrast the hazard rate, where negativity genuinely *is* an arbitrage. This asymmetry is reflected in the code: the credit curve enforces $h \geq 0$; the discount curve does not enforce $f \geq 0$.

3.2 Zero rates are D in rate units

A rate is meaningless without a compounding convention. Same $D(t)$, four legitimate answers:

Convention	Definition	Inverted
Continuous	$D = e^{-z_c t}$	$z_c = -\ln D/t$
Annual	$D = (1 + z_a)^{-t}$	$z_a = D^{-1/t} - 1$
m -periodic	$D = (1 + z_m/m)^{-mt}$	$z_m = m(D^{-1/(mt)} - 1)$
Simple	$D = (1 + z_s t)^{-1}$	$z_s = (1/D - 1)/t$

Linked by $1 + z_a = e^{z_c}$, with $z_m \rightarrow z_c$ as $m \rightarrow \infty$. The statement “the 5-year zero is 4.2%” is incomplete until the compounding *and* the day count are also stated. On the curve used later in this document, $D(5) = 0.82201223$ quotes as 3.9200% continuous, 3.9393% quarterly, 3.9587% semi-annual, 3.9978% annual, or 4.3305% simple — a 411bp spread across quotations of one identical number. The library stores $\ln D$ internally and converts on the way out, so there is a single representation and no convention drift.

3.3 Forward rates, derived by replication

You wish to lend from t_1 to t_2 at a rate agreed today. Two ways to reach a known amount at t_2 :

Strategy A. Buy $1/D(t_2)$ units of the t_2 zero. Cost today: 1. Payoff at t_2 : $1/D(t_2)$.

Strategy B. Buy $1/D(t_1)$ units of the t_1 zero. Cost today: 1. At t_1 receive $1/D(t_1)$ and roll to t_2 at a rate $f(t_1, t_2)$ agreed *today*. Payoff at t_2 : $\frac{1}{D(t_1)} e^{f(t_1, t_2)(t_2 - t_1)}$.

Both cost 1 today and pay an amount known today with certainty. Two riskless portfolios with identical cost must have identical payoff, else one shorts the rich and buys the cheap for a certain profit at zero net investment. Equating:

$$\frac{1}{D(t_2)} = \frac{1}{D(t_1)} e^{f(t_1, t_2)(t_2 - t_1)} \quad \implies \quad \boxed{f(t_1, t_2) = \frac{\ln D(t_1) - \ln D(t_2)}{t_2 - t_1}}$$

Under simple compounding — what an FRA or a SOFR-style coupon actually pays — the same argument gives

$$f_{\text{simple}}(t_1, t_2) = \frac{1}{\tau} \left(\frac{D(t_1)}{D(t_2)} - 1 \right)$$

where τ is the *contract's* accrual fraction under *its* day count, which is generally not $t_2 - t_1$ measured in curve time.

3.4 Instantaneous forwards: zeros are averages

Taking $t_2 \downarrow t_1$:

$$f(t) = -\frac{d}{dt} \ln D(t) \quad \Longleftrightarrow \quad D(t) = \exp\left(-\int_0^t f(s) ds\right),$$

the same structure as the hazard rate. Combining with $z_c(t) = -\ln D(t)/t$:

$$z_c(t) = \frac{1}{t} \int_0^t f(s) ds \quad (2)$$

Zero rates are running averages of instantaneous forwards. Forwards are the marginal quantity, zeros the cumulative one — so forwards are always more volatile, a humped forward curve produces a monotone zero curve, and an interpolation that looks smooth on zeros can be violent on forwards.

3.5 Why log-linear on discount factors

Interpolate linearly in $\ln D$. For $t \in [t_i, t_{i+1}]$ with $\theta = (t - t_i)/(t_{i+1} - t_i)$:

$$\ln D(t) = (1 - \theta) \ln D(t_i) + \theta \ln D(t_{i+1}) \quad \Longleftrightarrow \quad D(t) = D(t_i) \left(\frac{D(t_{i+1})}{D(t_i)} \right)^\theta$$

Four arguments, in ascending order of practical weight:

1. **Positivity is structural.** $D = \exp(\cdot) > 0$; the one genuine arbitrage condition cannot be violated by construction.
2. **It implies piecewise-constant instantaneous forwards.** Differentiating, $f(t)$ is the constant slope $\frac{\ln D(t_i) - \ln D(t_{i+1})}{t_{i+1} - t_i}$ on each bucket. Log-linear DF interpolation is therefore the *minimal-information* choice: the flattest forward curve consistent with the quotes, injecting no structure the market did not supply.
3. **It is internally consistent.** Any forward computed between two points inside one bucket returns that bucket's forward, regardless of sampling.
4. **It localises risk.** Bump the 5Y quote: only the two adjacent buckets' forwards move, so bucketed DV01 is clean. Under a cubic spline on zeros the perturbation propagates across the whole curve and hedge ratios smear into untouched tenors. This is the reason desks use it.

What the alternatives cost

- **Linear on D :** $f(t) = -D'/D = c/(a + bt)$, a hyperbola inside each bucket.
- **Linear on zero rates:** $f(t) = z(t) + t z'(t)$, giving a sawtooth forward curve with jumps at every knot. It can manufacture *negative forwards from strictly positive zero rates* whenever $z' < -z/t$, i.e. on any steep inversion.
- **Cubic spline on zeros:** smooth zeros, ringing forwards, globally non-local.

Measured on the curve used in this document, comparing instantaneous forwards:

t	log-linear DF	linear zeros
3.25	3.8150%	3.8675%
3.75	3.8150%	3.8325%
4.25	3.8150%	3.7975%
4.75	3.8150%	3.7625%
5.25	4.1300%	4.0850%
6.75	4.1300%	4.1750%

Linear zeros slope *down* through 3–5Y and *up* through 5–7Y. No quote said that.

The accepted cost

The forward curve under log-linear DFs is a step function, discontinuous at knots. This is irrelevant for discounting and linear products. If smooth forwards were required (convexity-sensitive exotics), monotone-convex or tension splines would be the move, at the price of losing property (2) and weakening (4).

3.6 Day counts, and where money actually leaks

Two separate jobs, routinely conflated:

1. **Curve time:** mapping dates to the t in $D(t)$. One convention throughout. This library uses ACT/365F, the ISDA CDS model convention.
2. **Accrual fractions:** computing τ for a specific contract's coupon, governed by that contract's terms.

Convention	τ	Where it lives
ACT/360	actual days / 360	USD & EUR money market, SOFR, CDS premium
30/360	$\frac{360\Delta y + 30\Delta m + \Delta d}{360}$	US corporates, USD fixed swap leg
ACT/365F	actual days / 365	GBP money market, ISDA CDS curve time
ACT/ACT	actual / actual	US Treasuries

The 360-day year. ACT/360 divides by 360 but counts real days, so a calendar year accrues $365/360 = 1.0139$ “years” and a rate quoted ACT/360 understates the annualised rate: $r_{365} = r_{360} \times \frac{365}{360}$. A 5.00% ACT/360 rate is 5.069% ACT/365 — 6.9bp of pure convention.

Concrete, on 10mm over one quarter, 15 Jun → 15 Sep (92 actual days), 5.00%:

ACT/360: \$127,777.78 30/360: \$125,000.00

\$2,777.78 difference. One leg, one quarter, same rate, same dates. Annualised across a swap book, convention errors are a live P&L break rather than a rounding issue.

3.7 Validation

Every compounding convention inverts its own definition to 10^{-14} ; $1 + z_a = e^{z_c}$ holds; $z_m \rightarrow z_c$ monotonically ($m = 4, 52, 365, 10^4$; error $< 10^{-5}$); the replication identity $D(t_1)e^{-f(t_1, t_2)(t_2 - t_1)} = D(t_2)$ holds to 10^{-14} , so the derivation is tested rather than assumed; $z_c(t) = \frac{1}{t} \int_0^t f$ verified by quadrature to 10^{-6} ; forwards constant inside a bucket to 10^{-13} .

QuantLib: `ql.DiscountCurve` is `InterpolatedDiscountCurve<LogLinear>`, so parity holds everywhere rather than only at knots — matched at 11 points from 0.07Y to 30Y at 10^{-12} relative,

plus `zeroRate` under both `Continuous` and `Compounded/Annual`, plus `forwardRate` at three intervals.

4 CDS Valuation

4.1 Setup and the one assumption that matters

Value at $t = 0$, notional N . The protection buyer pays a running coupon s on dates $t_1 < \dots < t_M$ with accrual fractions Δ_i (ACT/360). The seller pays $(1 - R)N$ at default time τ if $\tau \leq T$.

Default and interest rates are assumed independent. This lets $\mathbb{E}[D(\tau)\mathbf{1}_{\tau \leq T}]$ factor into $\int D(t)(-dQ(t))$ with D deterministic. It is market standard and it is *wrong in exactly the states one cares about* — in a credit crisis defaults cluster and rates rally, so $D(\tau)$ is high precisely when τ is small. The correction is second-order for investment-grade single names and material for distressed. The assumption is defensible; pretending it is costless is not.

4.2 Protection leg

The payoff is $(1 - R)$ at the random time τ , contingent on $\tau \leq T$:

$$\text{PV}_{\text{prot}} = (1 - R) \mathbb{E}[D(\tau)\mathbf{1}_{\tau \leq T}] = (1 - R) \int_0^T D(t) (-dQ(t)).$$

Here $-dQ(t)$ is the unconditional probability of defaulting in $[t, t + dt)$. Converting to hazard form using $Q = e^{-H}$, so $Q'(t) = -h(t)Q(t)$:

$$\text{PV}_{\text{prot}} = (1 - R) \int_0^T D(t) h(t) Q(t) dt \quad (3)$$

Read directly: $h(t) dt$ is the chance of defaulting in the next instant given survival, $Q(t)$ the chance of getting there, $D(t)$ discounts it.

Evaluating it exactly

Most implementations Riemann-sum this integral. They need not. On any interval $[a, b]$ where f and h are *both* constant — guaranteed on the merged grid of discount-curve and credit-curve knots — write $D(t)Q(t) = D(a)Q(a)e^{-(f+h)(t-a)}$ and set $c = f + h$:

$$\int_a^b D h Q dt = h D(a)Q(a) \int_0^{b-a} e^{-cu} du = \frac{h}{c} D(a)Q(a) [1 - e^{-c(b-a)}],$$

and since $D(b)Q(b) = D(a)Q(a)e^{-c(b-a)}$ this collapses to

$$\int_a^b D h Q dt = \frac{h}{f + h} [D(a)Q(a) - D(b)Q(b)] \quad (4)$$

Summing over sub-intervals gives an **exact** protection leg. The only error is the curve interpolation already chosen upstream — no quadrature error is layered on top. This is precisely why Steps 1 and 2 both used piecewise-flat rates.

Sanity check: setting $f = 0$ makes the factor 1 and the sum telescopes to $(1 - R)[Q(0) - Q(T)] = (1 - R)(1 - Q(T))$, the undiscounted expected loss.

4.3 Premium leg

Scheduled coupons

$$\text{PV}_{\text{coupon}} = s \sum_{i=1}^M \Delta_i D(t_i) Q(t_i)$$

Accrual on default — why the term exists

Premium accrues *continuously* but pays *quarterly*. If default lands at $\tau \in (t_{i-1}, t_i)$, the buyer has enjoyed protection from t_{i-1} to τ without paying for it, since the coupon was not yet due. The contract requires that stub at settlement.

Without this term, a CDS would be worth a different amount depending on *where in the coupon cycle it is traded*. Protection bought the day after a coupon date would carry up to 90 days of free protection in the default state; bought the day before, a full quarter's premium for one day of cover. The standardised 100bp/500bp fixed-coupon contract would not be fungible across trade dates. **The accrual term is what makes the instrument standardisable** — not a rounding refinement.

$$\text{PV}_{\text{accr}} = s \sum_i \frac{\Delta_i}{t_i - t_{i-1}} \int_{t_{i-1}}^{t_i} (t - t_{i-1}) D(t) h(t) Q(t) dt$$

The prefactor converts a curve-time offset into an accrual fraction; it is exact whenever the day count is proportional within a period, true for ACT/360 and ACT/365F up to day-rounding.

The same closed form is available. On $[a, b]$ inside a period starting at p , with $c = f + h$ and $L = b - a$, using $\int_0^L u e^{-cu} du = \frac{1 - e^{-cL}(1 + cL)}{c^2}$:

$$\int_a^b (t - p) DhQ dt = h D(a)Q(a) \left\{ \frac{1 - e^{-cL}(1 + cL)}{c^2} + (a - p) \frac{1 - e^{-cL}}{c} \right\}$$

4.4 The risky annuity and mark-to-market

$$A = \underbrace{\sum_i \Delta_i D(t_i) Q(t_i)}_{\text{coupons}} + \underbrace{\sum_i \frac{\Delta_i}{t_i - t_{i-1}} \int (t - t_{i-1}) DhQ dt}_{\text{accrual}}, \quad \text{PV}_{\text{prem}} = s \cdot A$$

A does not depend on s . For the protection buyer (long protection, short credit):

$$\text{MTM}_{\text{buyer}} = \text{PV}_{\text{prot}} - s \cdot A$$

Under the post-2009 standard contract s is fixed at 100bp or 500bp and the MTM is exchanged upfront, so this is literally the upfront payment.

4.5 Par spread: a ratio, not a solve

$$\text{MTM} = 0 \implies \boxed{s_{\text{par}} = \frac{\text{PV}_{\text{prot}}}{A}} \quad (5)$$

Because the protection leg does not involve s and A does not either, the par spread is a quotient — no root-finding. This matters precisely because the bootstrap *does* require a solve: there the unknown is the hazard rate, not the spread. It also gives

$$\text{MTM}_{\text{buyer}} = (s_{\text{par}} - s_{\text{contract}}) \cdot A,$$

from which CS01 follows.

4.6 Clean versus dirty par spread

Under the post-2009 standard the accrual period begins at the IMM date preceding the trade, so the buyer pays a *full* first coupon and is rebated the accrued portion. A par spread computed from the raw annuity therefore includes a stub of coupon for which no protection is provided.

The market convention is the **clean** par spread, which nets the accrued rebate out of the annuity. On a 1-year contract that stub can be 16% of the annuity. Bootstrapping from the dirty spread overstates the front hazard by roughly 18bp — a 20% error at the front of the curve. `par_spread(clean=True)` is the default.

4.7 The credit triangle, exactly

With flat h , flat r , and *continuously paid* premium:

$$A_{\text{cont}} = \int_0^T e^{-(r+h)t} dt = \frac{1 - e^{-(r+h)T}}{r + h}, \quad \text{PV}_{\text{prot}} = (1 - R)h \cdot \frac{1 - e^{-(r+h)T}}{r + h}$$

Dividing:

$$s_{\text{par}} = h(1 - R) \quad \text{exactly, independent of } r \text{ and } T \quad (6)$$

The credit triangle is not an approximation in this limit — it is an identity. The approximation error in practice comes entirely from *discrete* premium payment. This is the primary validation target for the whole pricer.

4.8 Risk measures — four different things

Measure	Definition	Note
RPV01 (A)	PV of 1 unit of spread, survival-contingent	Not a risk number; a <i>unit</i> . Quoted per unit notional.
CS01	Δ MTM for +1bp on the par spread curve, re-bootstrapped	$\approx A \cdot N \cdot 10^{-4}$ to first order. Second-order gap because moving spreads moves Q , which moves A .
IR DV01	Δ MTM for +1bp on the discount curve	Small: the legs largely offset.
JTD	$(1 - R)N - \text{accrued} - \text{MTM}$	Discontinuous. Not a derivative.

4.9 Hand-checked example

Flat $r = 5\%$ continuous, flat $h = 2\%$, $R = 40\%$, $T = 1$, one annual payment, no accrual-on-default. Every quantity is derivable on paper in three lines:

$$\text{PV}_{\text{prot}} = 0.60 \cdot \frac{0.02}{0.07} \cdot (1 - e^{-0.07}), \quad A = e^{-0.07}, \quad s_{\text{par}} = \frac{\text{PV}}{A}$$

	Engine		Hand	Difference
PV _{prot}	0.011589630873	0.011589630873		1.7×10^{-18}
RPV01	0.932393819906	0.932393819906		0
s_{par}	124.299739 bp	124.299739 bp		—

4.10 Convergence to the credit triangle

Target $h(1 - R) = 120\text{bp}$:

	Frequency	No accrual	With accrual
	1	124.2997 bp	123.0401 bp
	4	121.0562 bp	120.7525 bp
	52	120.0808 bp	120.0577 bp
	365	120.0117 bp	120.0083 bp

Monotone, and inside 0.01bp at daily frequency. Verified independent of r and T across nine (r, T) combinations.

What the accrual term actually does

A sharper claim than “it improves accuracy”. Over a period of length δ the discrete coupon undervalues the continuous one by $\approx (f + h)\frac{\delta^2}{2}DQ$, while the accrual integral contributes $\approx h\frac{\delta^2}{2}DQ$. It therefore cancels *exactly the hazard component* of the discretisation error and leaves the discounting component. Prediction: the residual error ratio tends to $f/(f + h) = 0.05/0.07 = 0.7143$. Measured at quarterly frequency: $0.7525/1.0562 = 0.7125$. This is now a test.

4.11 The 8bp gap, decomposed

A QuantLib comparison (flat 4%/3%, 5Y IMM) prices at 172.03bp, not the 180bp the credit triangle predicts. That gap is contract economics, not error, and factors cleanly:

1. **ACT/360 versus ACT/365F**. Every Δ_i is 365/360 too large relative to curve time, inflating the annuity. Measured ratio: 0.9863014 against $360/365 = 0.9863014$ — nine digits.
2. **Seasoned stub**. Accrual began 20 Jun; protection runs from 17 Aug. The annuity carries 58/360 of a coupon buying no protection — 3.59% of RPV01.

$$180.90 \times \frac{360}{365} \times (1 - 0.0359) = 172.014\text{bp} \quad \text{versus} \quad 172.027\text{bp priced}$$

A residual of 0.013bp. Trading the same maturity *on* an IMM date recovers 178.42bp: wedge 2 vanishes, wedge 1 remains.

4.12 Worked example: 5Y, 10mm notional, 100bp coupon

Protection leg PV	390,711.35
Premium leg PV	466,061.41
coupons	465,222.72
accrual	838.69
RPV01 (per unit)	4,660.614
MTM	−75,350.06
Par spread	83.8326 bp
Accrued (rebate)	16,111.11
Upfront points	−0.7535%
CS01 (reprice)	4,689.72
CS01 (analytic)	4,660.61 (−0.62%)
IR DV01	6.54 (0.14% of CS01)
Rec01 (+1pt)	−6,511.86
JTD	6,059,238.95 (1,292× CS01)

Buying 100bp protection on an 84bp name loses 75k, as it should. Accrual is 0.18% of the premium leg in PV — nearly nothing — while moving the par spread meaningfully. IR DV01 is 0.14% of CS01: the legs almost fully offset.

JTD is 1,292× CS01. A book can be flat CS01 and still be destroyed by a single default. That ratio is why desks limit both, and why an index hedge that neutralises spread risk leaves idiosyncratic default risk entirely untouched.

5 Bootstrapping the Survival Curve

5.1 The inverse problem is underdetermined until parameterised

Forward: given $h(\cdot)$, compute $s_{\text{par}}(T)$. Backward: given n quotes, find $h(\cdot)$. Infinitely many hazard functions reproduce n quotes.

The market determines n numbers, not a function. The bootstrapped curve is *quotes plus an interpolation assumption*, never “the market’s view of default”.

Piecewise-constant h with knots placed *exactly at the quote maturities* gives n unknowns for n equations: square system, unique solution, zero residual. Knots anywhere else turn this into least-squares fitting, a different problem with different failure modes.

5.2 Why sequential: the Jacobian is triangular

Define the residual $\Phi_k = \text{PV}_{\text{prot},k} - s_k A_k$. The k -th CDS matures at T_k , so both legs integrate against $Q(t)$ only for $t \leq T_k$, and Q on that range depends only on $h_1, \dots, h_k$. Hence

$$\frac{\partial \Phi_k}{\partial h_j} = 0 \quad \text{for all } j > k, \quad J = \begin{pmatrix} * & 0 & 0 & 0 & 0 \\ * & * & 0 & 0 & 0 \\ * & * & * & 0 & 0 \\ * & * & * & * & 0 \\ * & * & * & * & * \end{pmatrix}$$

Lower triangular, so the solution is forward substitution: an n -dimensional nonlinear system collapses to n one-dimensional root-finds. Four consequences, in ascending order of practical weight:

1. **Robustness.** Bracketed 1-D root-finding on a monotone function essentially always converges. Multivariate Newton needs a good starting point and can diverge into negative hazards.
2. **Exact reproduction.** n equations, n unknowns — the fit is exact, not least-squares.
3. **Economic meaning.** h_k is the *marginal* default intensity over $(T_{k-1}, T_k]$ implied by the incremental information in the k -th quote. It is a forward quantity, like a forward rate. The 5Y quote does not describe aggregate 0–5Y default risk; it describes 3–5Y risk given what 1Y and 3Y already said.
4. **Error localisation.** A bad 10Y quote cannot corrupt h_1 . Errors propagate forward, never backward.

5.3 Why solve Φ rather than the par-spread residual

Both have the same root and the same monotonicity, but the ratio form degenerates: at the first tenor $A \rightarrow 0$ as h grows, so s_{par} is unbounded (reaching 3.01 at $h = 5$ on a 1Y contract) while Φ stays bounded within $[-0.012, 0.589]$ and is measurably less curved. Both failure modes remain detectable in the Φ form: $\Phi_k(0) > 0$ catches the inverted-curve case and bracket-expansion failure catches the ceiling.

5.4 Why each 1-D solve is well posed

Raising h_k raises PV_{prot} (the contribution over $(T_{k-1}, T_k]$ rises monotonically toward the certain-default limit $D(T_{k-1})Q(T_{k-1})$) and lowers the coupon part of A (every $Q(t_i)$ for $t_i > T_{k-1}$ falls). Both push Φ the same way, so Φ is strictly increasing and has at most one root.

One honest caveat: the *accrual* part of A also rises with h , so denominator monotonicity is not unconditional. That term is $O(h\delta^2)$ and is dominated over any realistic range. It is verified numerically over $h \in [10^{-6}, 5]$ rather than asserted as a theorem.

5.5 Failure modes fall out of the bracketing

Spread above the certain-default ceiling

As $h_k \rightarrow \infty$, default becomes certain immediately after T_{k-1} :

$$s^{\max}(T_k) = \frac{\text{PV}_{\text{prot}}(T_{k-1}) + (1 - R) D(T_{k-1}) Q(T_{k-1})}{A(T_{k-1}) + \varepsilon_{\text{accr}}}$$

Finite for every $k \geq 2$ (unbounded for $k = 1$, where $A \rightarrow 0$). A quote above $s^{\max}$ cannot be reproduced by any non-negative hazard: it prices protection above the maximum possible loss given the earlier quotes. That is an arbitrage in the quote set, and the bootstrap must say so rather than silently returning garbage.

Negative implied forward hazard on an inverted curve

Inversion alone is normal — distressed names invert routinely, because near-term default risk dominates. What fails is inversion steep enough that reproducing $s_k < s_{k-1}$ requires $h_k < 0$, which means Q *rises* over $(T_{k-1}, T_k]$: a negative forward default probability. The arbitrage is to sell T_{k-1} protection and buy T_k protection, being paid to own a strictly non-negative claim over the window.

A rough threshold from the credit triangle: the implied forward hazard turns negative when $s_k T_k \lesssim s_{k-1} T_{k-1}$, i.e. when the spread $\times$ maturity product falls.

Recovery inconsistency and the upfront problem

R is an assumption, not a quote. Pushing R toward 1 collapses loss given default, so $s^{\max}$ falls and previously fine quotes become unbootstrappable. Distressed names are the acute case: the market stops quoting running spread and quotes *points upfront against a fixed coupon*, precisely because the par-spread representation degenerates. Bootstrapping a 2000bp name from a “par spread” is a modelling error before it is a numerical one.

5.6 Results

Quotes at 1/3/5/7/10Y of 55/82/105/118/130bp, $R = 40\%$:

Tenor	Quoted	Repriced	Error	Hazard	Fwd spread	Survival
1.0	55.0000	55.0000	-2.6×10^{-14}	92.35	55.41	0.992263
3.0	82.0000	82.0000	-1.7×10^{-14}	158.48	95.09	0.961264
5.0	105.0000	105.0000	$+3.5 \times 10^{-14}$	239.62	143.77	0.916283
7.0	118.0000	118.0000	-5.2×10^{-14}	263.71	158.22	0.869146
10.0	130.0000	130.0000	$+1.7 \times 10^{-14}$	282.38	169.43	0.798488

Maximum repricing error: 4.44×10^{-16} relative — two units in the last place of a float64. Verified two independent ways: par spread equality, and pricing each quoted CDS at its own coupon off the bootstrapped curve ($|\text{MTM}| < 10^{-7}$ on 10mm).

Note the hazard column against the forward-spread column. The 5Y quote is 105bp, but the marginal intensity over 3–5Y implies a 143.77bp forward spread. **Bootstrapped hazards are forward quantities.** Reading h_5 as “the 5Y default intensity” is the standard error.

5.7 Triangularity, made testable

`test_locality_of_the_triangular_system` bumps the 10Y quote by 50bp and asserts $h_1..h_4$ are unchanged using `np.array_equal` — bit-for-bit, not approximately. `test_subset_bootstrap_matches_prefix` bootstraps only the first three quotes and asserts exact equality with the first three hazards of the full run. That *is* forward substitution.

5.8 Feasible quote windows

Tenor	Floor	Ceiling	Quoted
1Y	0.00 bp	119,429.26 bp	55.00
3Y	16.95 bp	6,576.34 bp	82.00
5Y	50.36 bp	1,985.56 bp	105.00
7Y	78.17 bp	1,154.55 bp	118.00
10Y	88.80 bp	807.05 bp	130.00

The floor rises and the ceiling collapses moving out the curve: each new quote is constrained on both sides by everything preceding it. The 1Y has no floor (nothing precedes it) and an effectively unbounded ceiling.

5.9 The recovery degeneracy

Same quotes, two recovery assumptions, both repricing to 4×10^{-16} :

$$\frac{h(R = 40\%)}{h(R = 20\%)} = [1.333, 1.334, 1.337, 1.340, 1.343] \quad \text{against} \quad \frac{0.8}{0.6} = 1.333$$

Implied 5Y cumulative default probability: 8.37% at $R = 40\%$, 6.34% at $R = 20\%$.

CDS spreads identify expected loss $h(1 - R)$, not h . A two-point spread in implied default probability from an assumption nobody observes. Any sentence beginning “the CDS market is pricing an $X\%$ chance of default” is really “... given someone’s recovery assumption”.

6 Application: Five U.S. Bank Holding Companies

6.1 What this analysis can and cannot claim

One quote per name is not a term structure. Each name supplies a single 5Y spread, so the bootstrap has one degree of freedom and returns a *flat* hazard. Every point away from 5Y is the piecewise-flat assumption, not market information. The chart marks the 5Y point as observed and renders everything else dashed. Obtaining 1/3/5/7/10Y quotes per name would produce a genuine term structure — the engine already handles it; the constraint is data, not model.

Two further constraints carried into every conclusion below:

- **Risk-neutral, not actuarial.** These probabilities embed a default risk premium, a liquidity premium, and correlation compensation. They run roughly $8\text{--}12\times$ historical default rates for comparable ratings.
- **Indicative spreads, not executable marks.** Post-2014 single-name U.S. bank CDS is materially less liquid than pre-crisis, with much flow in CDX.NA.IG.

6.2 Why 40% recovery, and why it is the weakest input

40% is the **ISDA quoting convention** for senior unsecured obligations of developed-market corporates. It makes quotes comparable. It is *not* an estimate of what these instruments would recover. For U.S. bank holding companies the convention is doing more work than usual:

1. **Holdco senior is the bail-in layer.** Single-name CDS on U.S. banks references the *holding company*. Under Dodd-Frank Title II Orderly Liquidation Authority and the single-point-of-entry resolution strategy, holdco senior unsecured is explicitly TLAC — it exists to be written down or converted so the operating subsidiaries continue. Its recovery in the scenario that triggers a CDS credit event is by design low, not the corporate-average 40%.
2. **The one clean data point is 8.625%.** Lehman Brothers senior CDS settled at 8.625 cents in the October 2008 auction — roughly a fifth of the convention.
3. **The empirical base is thin.** Bank senior recoveries are governed by resolution mechanics never tested at scale on a U.S. G-SIB.

Practically: 40% is defensible *as a convention* and is the right choice for consistency with how the quotes were formed. It is not defensible as a forecast.

6.3 Results

Name	5Y spread	Hazard	1Y PD	5Y PD	10Y PD
JPMorgan	58 bp	96.2 bp	0.96%	4.70%	9.17%
Bank of America	63 bp	104.5 bp	1.04%	5.09%	9.92%
U.S. Bancorp	67 bp	111.1 bp	1.11%	5.41%	10.52%
Wells Fargo	72 bp	119.4 bp	1.19%	5.80%	11.26%
Citigroup	85 bp	141.0 bp	1.40%	6.81%	13.15%

6.4 Three structural findings

The recovery assumption cannot touch the ranking

With a uniform R across names, $h = s/(1 - R)$ is a monotone transform of the spread, so the ordering is invariant — recovery moves levels only. Reordering requires *name-specific* recovery,

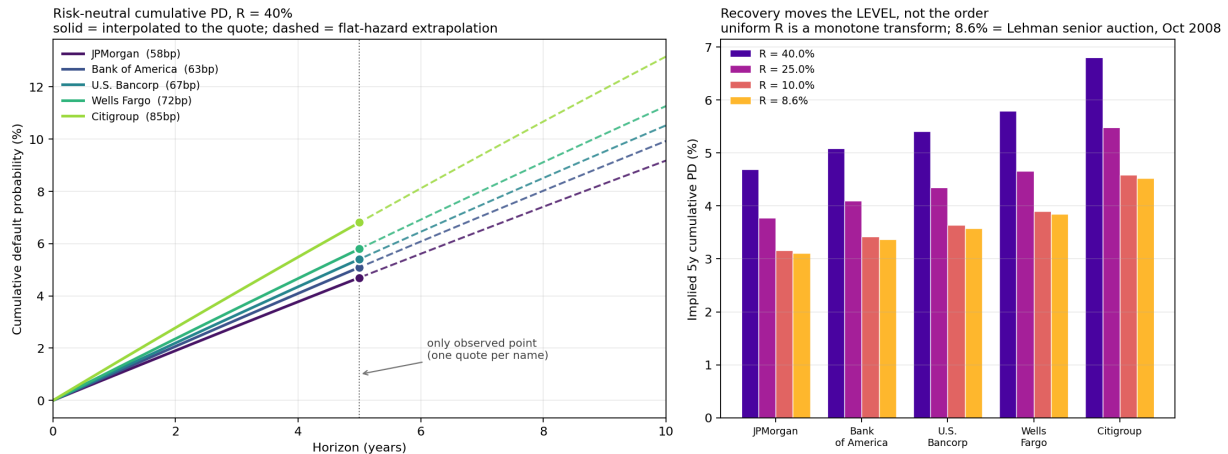


Figure 1: Left: risk-neutral cumulative default probability at $R = 40\%$. Solid to the quote, dashed beyond — the dashed portion is the flat-hazard assumption, not market data. Right: implied 5Y PD under four recovery assumptions.

and the required differential is large:

Ties JPMorgan at $R = 40\%$ if its own recovery is	
Bank of America	34.83%
U.S. Bancorp	30.69%
Wells Fargo	25.52%
Citigroup	12.07%

Citigroup only ties JPMorgan under a 28-point recovery differential. **State the ordering with confidence; state the levels with a recovery caveat attached.**

Rates are irrelevant here

A 600bp swing in the discount curve moves Citigroup's implied 5Y PD from 6.8300% to 6.7807% — five basis points of probability. The Step 4 IR DV01 result, confirmed end to end.

The entire cross-sectional signal is 27bp

The three tightest names sit inside 9bp of each other; JPM→BAC is 0.394pp of PD on a 5bp spread gap. Whether JPM, BAC and USB are genuinely distinguishable on this data is a question about bid/ask, not about credit.

6.5 Risk-neutral versus actuarial

Aa-rated historical 5Y cumulative default rate	0.31%
A-rated historical 5Y cumulative default rate	0.56%
Baa-rated historical 5Y cumulative default rate	1.55%
Market-implied range	4.70% – 6.81%
Ratio to A-rated actuarial	8.4× – 12.2×

Roughly an order of magnitude above realised default experience. That wedge is the risk premium — compensation for jump-to-default, illiquidity, and the fact that bank defaults arrive correlated with everything else in a portfolio. It is not a forecast error; it is why risk-neutral PD is a *pricing* quantity rather than an *actuarial* one.

6.6 Market-implied ranking against fundamental ranking

Where they agree: Citigroup

The market puts Citi 27bp wide of JPMorgan — a 47% spread premium and the only gap in the set that clears any plausible bid/ask. A thin-PPNR-coverage thesis and the market’s ordering point the same direction.

Precision matters about what that agreement is worth: Citi’s spread is also consistent with lower through-cycle ROTCE, ongoing transformation and consent-order costs, and a larger EM footprint. Concluding “the market is pricing my PPNR thesis” is over-attribution. The defensible claim is narrower — *the market independently ranks Citi last, and PPNR coverage is one of a small number of explanations consistent with that.*

Where they disagree: Bank of America

A held-to-maturity tail thesis says BAC carries the largest unrealised securities loss position in the peer set relative to tangible capital. The market prices BAC as the *second-safest* name — 5bp inside U.S. Bancorp and 9bp inside Wells Fargo. On this data the market does not price the HTM tail as a credit event driver at all.

The reframe

The instinct is to argue the market is wrong. The stronger move is to notice that CDS and a fundamental screen are answering *different questions*, and that both can be right.

A CDS prices the probability and severity of a *credit event*. HTM losses do not cause a credit event through the income statement. They cause one only through a **funding channel**: forced liquidation of the HTM book to meet outflows, crystallising the mark and destroying capital. That is the SVB mechanism, and it is a liquidity-plus-concentration failure, not a securities-portfolio failure.

For BAC that channel is weak: a very large, granular, heavily insured retail deposit base, a G-SIB LCR and NSFR regime, discount window access, and a post-March-2023 policy precedent for term lending against par. The market appears to be saying the HTM position is a *book value and NII problem, not a solvency problem* — that it suppresses tangible book, AOCI-adjusted CET1 and forward earnings power without moving the probability of resolution.

If that is right, the HTM thesis is not contradicted by 63bp — it is **mispriced in a different instrument**. A view that BAC’s HTM tail impairs earnings power and capital return is an *equity* view, or a preferred/AT1 view. Senior CDS is the wrong expression vehicle, and the fact that CDS does not reflect it is evidence about instrument selection rather than evidence against the thesis.

“The market and I are not disagreeing about Bank of America. We are pricing different questions — and my HTM thesis belongs in the capital structure below senior, where the loss actually lands.”

The falsifiable version

This framing generates a test rather than an opinion. If the HTM tail is genuine credit risk that senior CDS is missing, it should appear first in funding-sensitive instruments: BAC’s preferred/AT1 spreads widening relative to peers, the senior-sub CDS basis widening, or deposit beta and uninsured-deposit share deteriorating faster than the peer set. If those are flat while the equity discount persists, the market’s separation is correct and the thesis is an earnings thesis wearing credit clothing.

6.7 Highest-value extensions

1. **Obtain a term structure.** 1/3/5/7/10Y quotes per name turn five numbers into five curves, and the *shape* is where a funding thesis would show up — a concern that is real but slow should steepen the front end relative to peers. A single 5Y point structurally cannot express that.
2. **Add the sub-senior structure.** Senior/sub CDS basis and preferred spreads make the reframe above testable rather than rhetorical.
3. **Pull current holdco ratings** as an independent third ranking. A market/fundamental/agency three-way comparison is materially more persuasive than a two-way.

7 Validation Summary

86 tests, all passing. QuantLib is used only as an independent oracle.

Module	Tests	Coverage
<code>curves.py</code>	15	Analytic identities, no-arbitrage rejection, QuantLib <code>HazardRateCurve</code> parity at 10^{-12}
<code>discount_curve.py</code>	17	Convention inversions, replication identity, bucket-constant forwards, QuantLib <code>DiscountCurve</code> parity at 10^{-12}
<code>cds.py</code> + <code>schedule.py</code>	33	Two hand-checked closed forms, credit-triangle convergence, exact-vs-quadrature, QuantLib <code>MidPointCdsEngine</code> within 1bp
<code>bootstrap.py</code>	21	Machine-precision repricing, triangularity, failure modes, recovery degeneracy, QuantLib <code>PiecewiseFlatHazardRate</code> parity

7.1 Selected assertions

- **Exact integration versus brute force.** The closed-form protection integral is compared against a Riemann sum at grid sizes 20k, 200k and 2M. Because the integrand is discontinuous at every hazard knot, trapezoid converges slowly — so the test asserts *convergence toward* the closed form rather than agreement at one grid size. That is the correct direction of the claim: the closed form is exact and the Riemann sum is the approximation.
- **Exact versus midpoint.** The two methods agree to 0.0023bp on a 5Y par spread and are asserted *not* identical — if they were, one implementation would be wrong.
- **Locality.** `np.array_equal` on hazards after a 50bp bump to the 10Y quote. Bit-for-bit, not approximate.

7.2 Defects found and corrected

Documented because the process is part of the artifact.

1. **Inverted accrual direction.** An assertion that ACT/360 produces a higher quoted rate than ACT/365F failed. ACT/360 yields the *larger* year fraction ($91/360 = 0.25278$ against 0.24932), so for identical growth the quoted simple rate is *lower*. “Divide by 360” feels as though it should shrink the fraction. It does not.
2. **RPV01 scaled by notional.** The risky annuity was being reported multiplied by notional rather than per unit. Mathematically harmless (it cancels in the par spread) but against market convention and confusing on output. Corrected to the per-unit figure.
3. **Wrong maturity convention.** Schedules measured tenor from the *trade date*, making every contract up to a full quarter too long — a “5Y” maturing 20 Sep 2031 against QuantLib’s 20 Jun 2031. The CDS2015 convention measures tenor from the *accrual start* (the preceding IMM date). This produced a 66bp error in the bootstrapped 1Y hazard.
4. **Bootstrapping off the dirty par spread.** On a seasoned schedule the annuity carries a coupon stub buying no protection. On a 0.84Y contract that stub is 16% of the annuity, inflating h_1 by 18bp — 20% wrong at the front of the curve. Netting the accrued rebate collapsed the QuantLib gap:

	Dirty	Clean	QuantLib	Clean – QL
1Y	110.38	92.44	92.14	+0.31 bp
3Y	162.46	158.44	158.47	–0.03 bp
10Y	284.22	282.31	282.28	+0.04 bp

5. **Closure mutation in the bootstrap loop.** The residual function closed over the hazard list, which was appended to before the diagnostic re-evaluation — lengthening the array out from under the closure. Caught by 19 simultaneous test failures.
6. **Clean par spread against dirty mark-to-market.** Introducing the clean par spread (defect 4) netted the accrued rebate out of the annuity for s_{par} but not for the MTM, which continued to use the dirty annuity. On a seasoned schedule the identity $\text{MTM} = (s_{\text{par}} - s) A N$ therefore failed by exactly the accrued rebate, and pricing *at* the par spread returned –accrued rather than zero. Invisible to the suite because every MTM test used the synthetic `flat_schedule`, where `accrued_fraction()` is zero and the two conventions coincide. Surfaced by a user running the engine on a seasoned IMM schedule, not by inspection. Corrected by making `mtm` the clean market value, exposing `stream_pv` for the raw cashflow view (which jump-to-default correctly nets against, since on default the stub is owed rather than rebated), and adding seven regression tests on a deliberately seasoned fixture.
7. **Incorrect claim about recovery sensitivity.** A draft chart asserted that the recovery assumption drove the *ranking*. It does not: with uniform R , $h = s/(1 - R)$ is a monotone transform of the spread and the ordering is invariant. Recovery drives the *level*. Caught by running the sensitivity rather than reasoning about it.

Defects 3, 4 and 6 were each surfaced by a validation step, not by inspection. Defect 4 was diagnosed in a single step because the Step 4 day-count decomposition had already isolated the accrued-rebate wedge as a named quantity.

8 Limitations and Roadmap

8.1 Known simplifications

1. **No business-day adjustment.** Payment dates should roll Following against a real holiday calendar. This moves a handful of cashflows by 1–3 days and is the residual in the QuantLib comparisons ($\sim 0.3\text{bp}$ at 1Y, under 0.07bp beyond). Mechanical to add; omitted as a scoping choice and stated in the module docstring so the omission is auditable.
2. **Rates and default assumed independent.** Standard, and wrong in stress. Material for distressed names.
3. **Deterministic recovery.** R is a fixed input, not a distribution, and no recovery/default correlation is modelled.
4. **Running-spread quotation only.** The post-2009 standard contract quotes points upfront against a fixed 100bp or 500bp coupon. For distressed names the par-spread representation degenerates as the annuity collapses. An upfront bootstrap is the natural next addition.
5. **No curve-shape alternatives.** Piecewise-flat hazards are the only parameterisation. The bootstrapped curve is quotes *plus* that assumption.

8.2 Roadmap

1. Points-upfront quotation and conversion to and from par spread.
2. Bucketed CS01 computed off a full re-bootstrap, exploiting the locality property.
3. Business-day adjustment against a real calendar.
4. Senior/subordinated basis, for the capital-structure analysis in Section 6.
5. Index (CDX) decomposition against single-name constituents.

8.3 Reproducing this document

```
pip install numpy scipy pandas matplotlib pytest QuantLib
PYTHONPATH=. python -m pytest tests/ -q # 79 passed
jupyter notebook notebooks/bank_credit_analysis.ipynb
```

All figures in this document are produced by the code in the repository. Bank CDS spreads are indicative and supplied as inputs; the discount curve is illustrative. Historical default rates should be verified against the current Moody's Annual Default Study before external citation. Nothing here is investment advice.